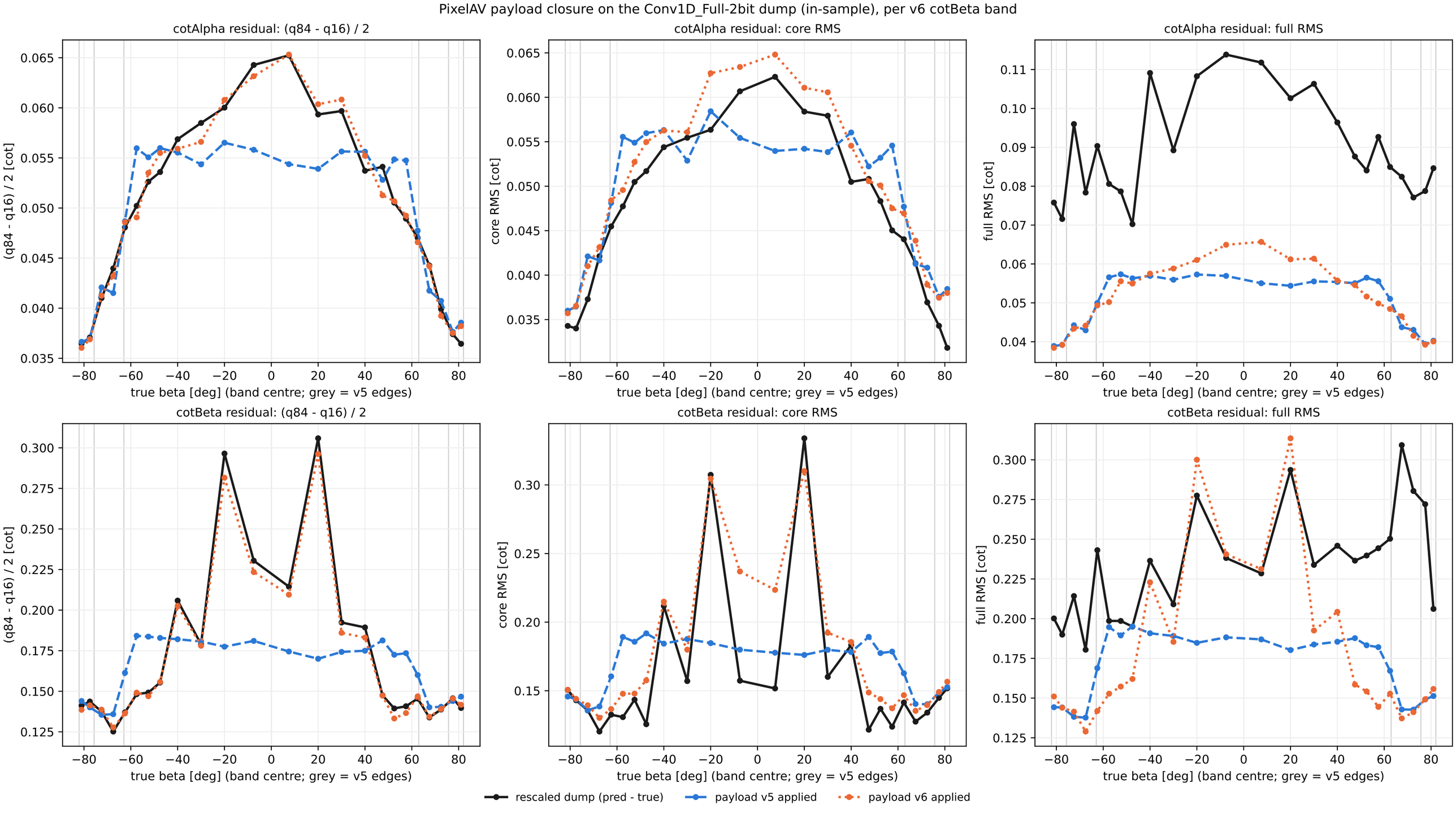
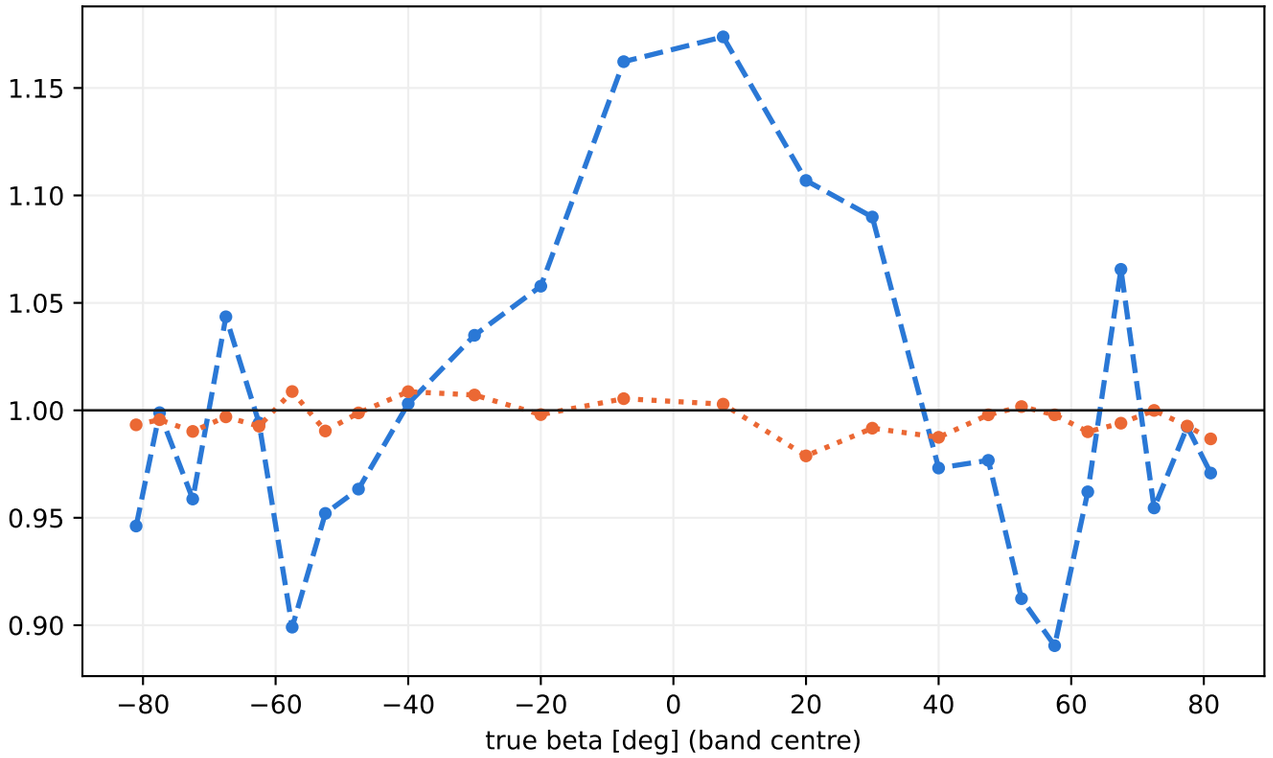
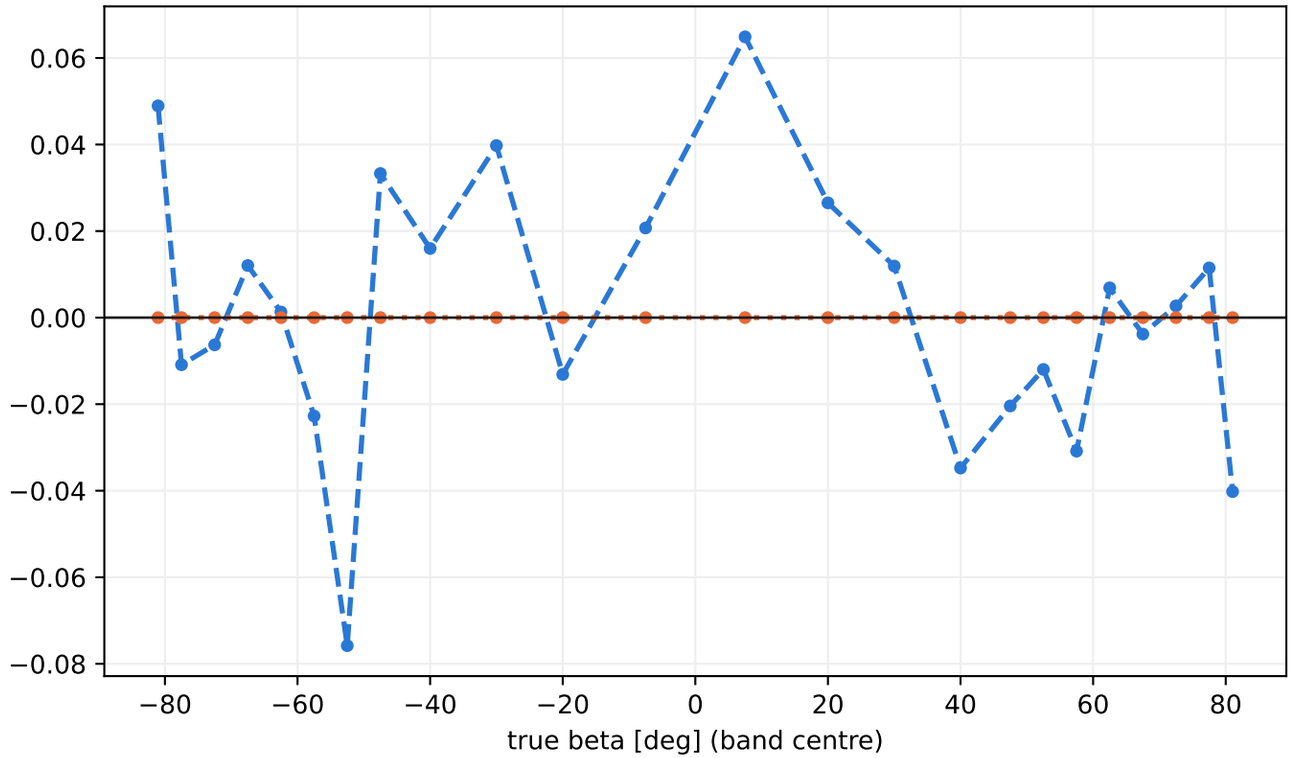

PixelAV payload closure on the Conv1D_Full-2bit dump (in-sample), per v6 cotBeta band
cotAlpha residual: (q84 - q16) / 2
cotAlpha residual: core RMS
cotAlpha residual: full RMS
cotBeta residual: (q84 - q16) / 2
cotBeta residual: core RMS
cotBeta residual: full RMS
(q84 - q16) / 2 [cot]
core RMS [cot]
full RMS [cot]
true beta [deg] (band centre; grey = v5 edges)
rescaled dump (pred - true)
payload v5 applied
payload v6 applied

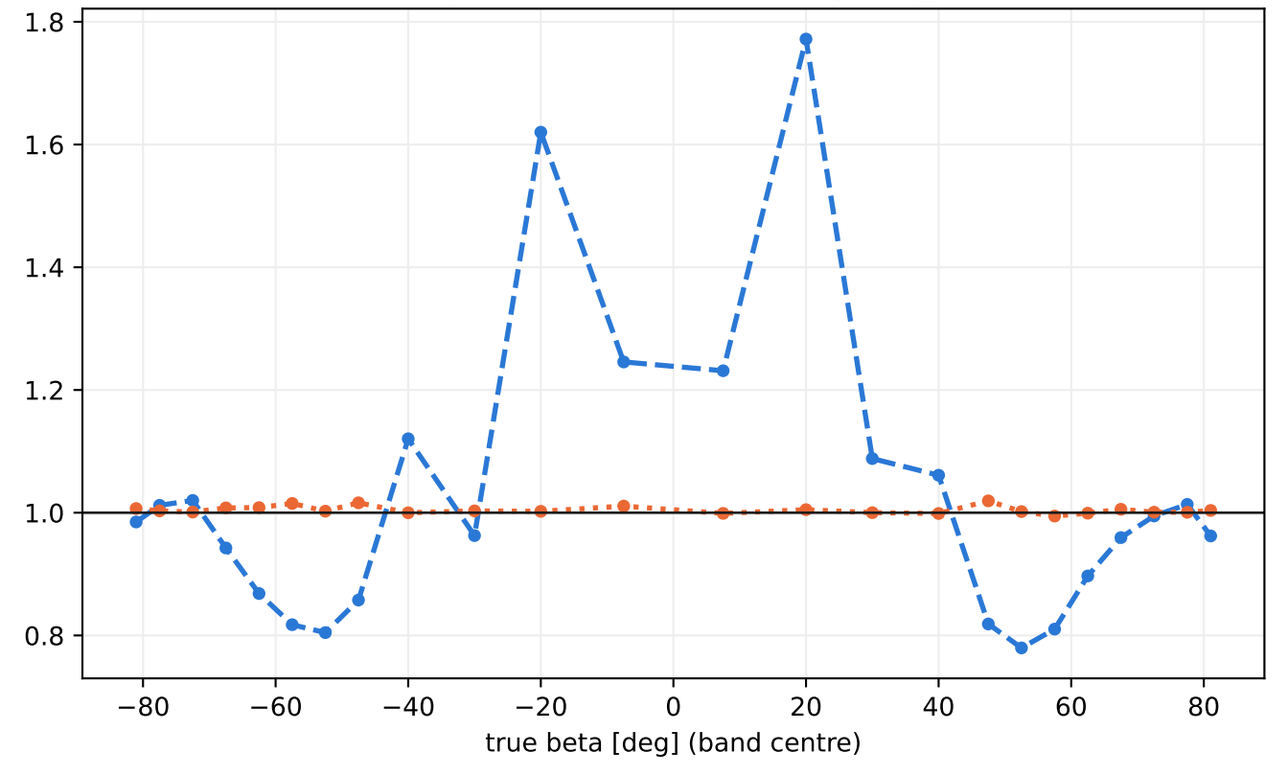
cotAlpha pull (dump residual - bias) / sigma: q_width



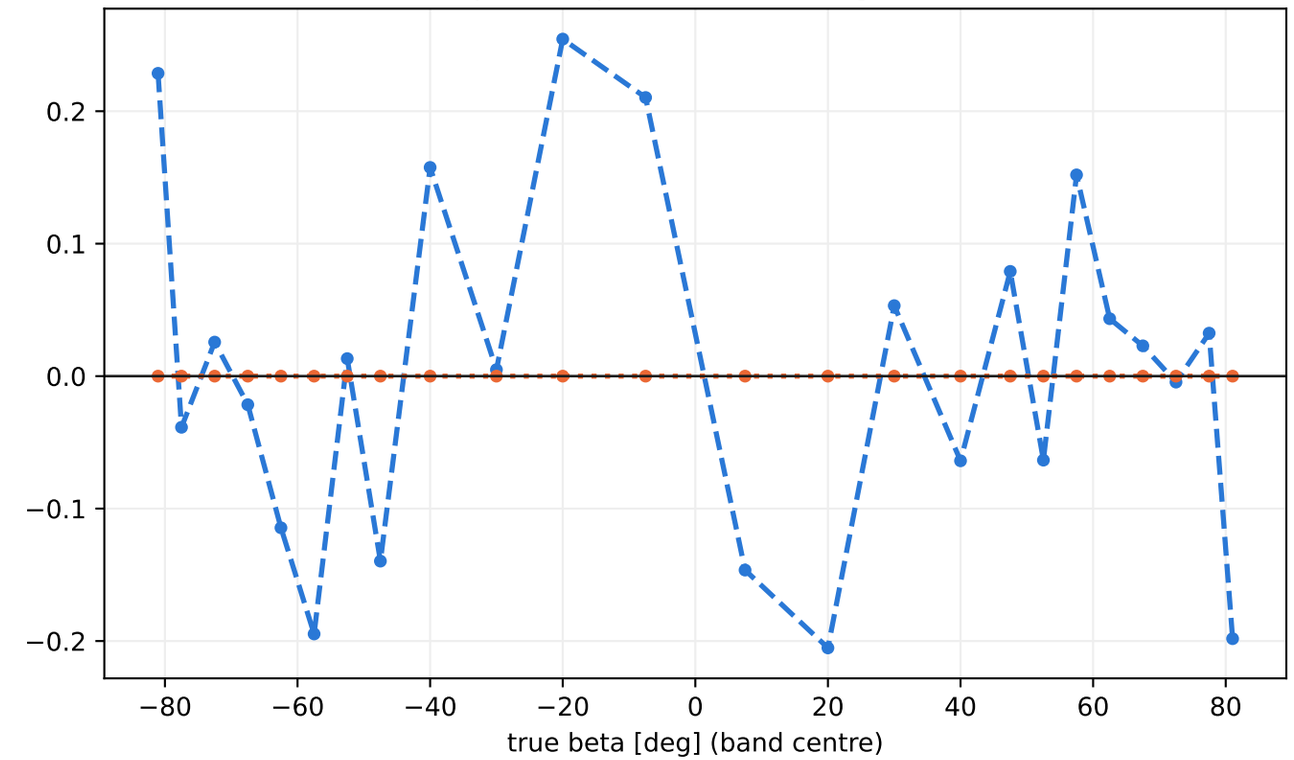
cotAlpha pull (dump residual - bias) / sigma: median



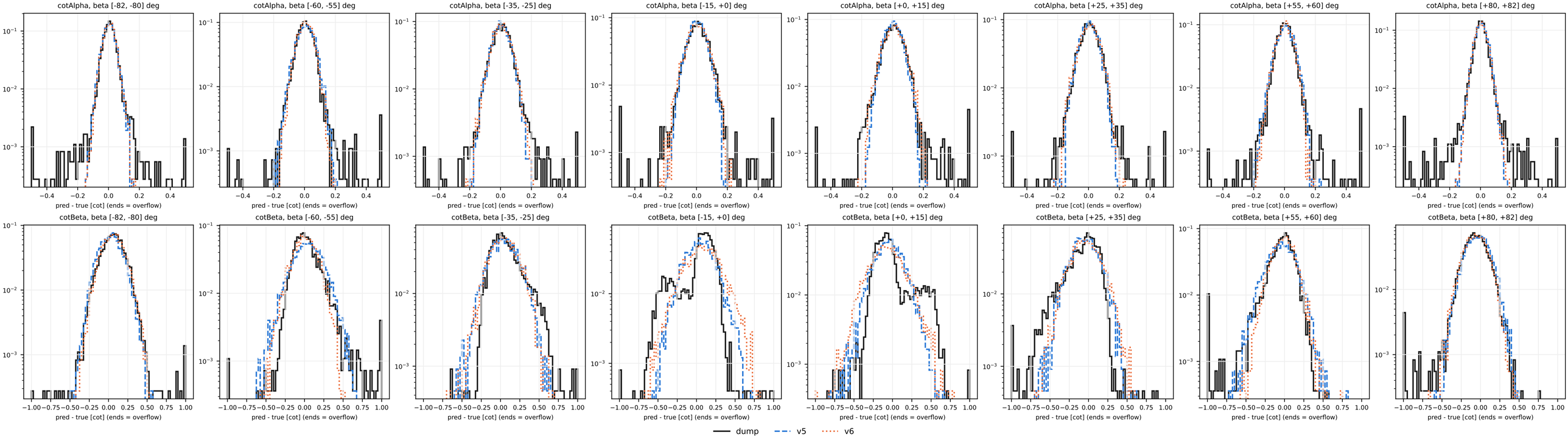
cotBeta pull (dump residual - bias) / sigma: q_width

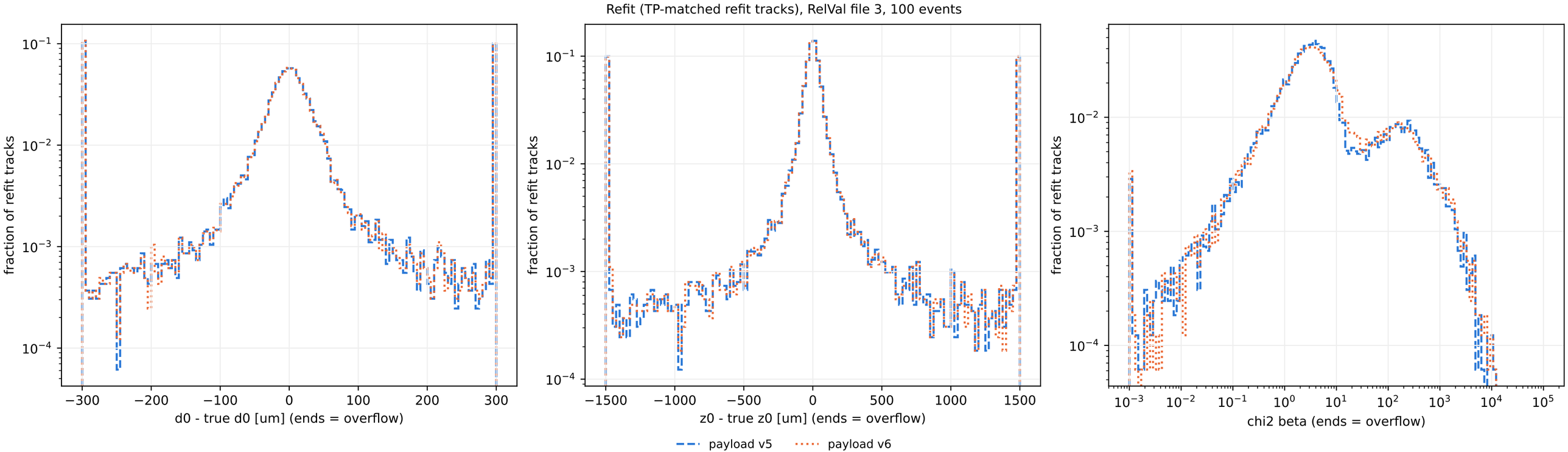


cotBeta pull (dump residual - bias) / sigma: median



payload v5 payload v6 ideal


cotAlpha, beta [-82, -80] deg
cotAlpha, beta [-60, -55] deg
cotAlpha, beta [-35, -25] deg
cotAlpha, beta [-15, +0] deg
cotAlpha, beta [+0, +15] deg
cotAlpha, beta [+25, +35] deg
cotAlpha, beta [+55, +60] deg
cotAlpha, beta [+80, +82] deg
cotBeta, beta [-82, -80] deg
cotBeta, beta [-60, -55] deg
cotBeta, beta [-35, -25] deg
cotBeta, beta [-15, +0] deg
cotBeta, beta [+0, +15] deg
cotBeta, beta [+25, +35] deg
cotBeta, beta [+55, +60] deg
cotBeta, beta [+80, +82] deg
pred - true [cot] (ends = overflow)
dump
v5
v6


Refit (TP-matched refit tracks), RelVal file 3, 100 events
fraction of refit tracks
d0 - true d0 [um] (ends = overflow)
z0 - true z0 [um] (ends = overflow)
chi2 beta (ends = overflow)
payload v5
payload v6

Dump closure per cotBeta band [cot]: q-width, core RMS, full RMS, median (1)

angle	beta_band_deg	N	dump_q_width	v5_q_width	v6_q_width	dump_core_rms	v5_core_rms	v6_core_rms	dump_full_rms	v5_full_rms	v6_full_rms	dump_median	v5_median	v6_median
alpha	[-82, -80]	3635	0.03646	0.03664	0.03605	0.03429	0.036	0.03571	0.07579	0.0389	0.03843	0.006039	0.003798	0.005218
alpha	[-80, -75]	16269	0.03708	0.03695	0.03689	0.034	0.03646	0.03654	0.07157	0.03921	0.03918	0.003946	0.004299	0.003952
alpha	[-75, -70]	8905	0.04101	0.04208	0.04125	0.03729	0.04211	0.04101	0.09599	0.04423	0.04338	0.003951	0.005281	0.005045
alpha	[-70, -65]	5574	0.04397	0.04151	0.04321	0.04213	0.04166	0.04314	0.07836	0.04294	0.04414	0.004849	0.003235	0.004085
alpha	[-65, -60]	3715	0.04807	0.04868	0.04858	0.04548	0.04812	0.04841	0.09034	0.04995	0.04934	0.004261	0.003808	0.003473
alpha	[-60, -55]	2763	0.05022	0.05596	0.04907	0.04773	0.05555	0.04957	0.08059	0.05657	0.0502	0.00367	0.008239	0.006535
alpha	[-55, -50]	2220	0.05263	0.05506	0.05351	0.05048	0.05489	0.05272	0.07866	0.05735	0.05555	0.002373	0.008156	0.00421
alpha	[-50, -45]	1742	0.05359	0.05598	0.05549	0.0517	0.05596	0.05497	0.07025	0.0563	0.05496	0.006612	0.004179	0.006211
alpha	[-45, -35]	2819	0.05687	0.05554	0.05592	0.05439	0.05632	0.05626	0.1091	0.05693	0.05752	0.005149	0.006024	0.006476
alpha	[-35, -25]	2191	0.05849	0.05436	0.05661	0.05545	0.05288	0.05608	0.08923	0.05595	0.05883	0.00635	0.002726	0.006046
alpha	[-25, -15]	1871	0.06002	0.05652	0.06079	0.05635	0.05843	0.06271	0.1083	0.05729	0.06102	0.003748	0.003488	0.003186
alpha	[-15, +0]	2489	0.06428	0.05581	0.06316	0.06068	0.05544	0.06342	0.1139	0.05693	0.06494	0.008098	0.003411	0.004827
alpha	[+0, +15]	2438	0.06524	0.05438	0.06531	0.06231	0.05396	0.06481	0.1118	0.05505	0.06568	0.008617	0.004602	0.008194
alpha	[+15, +25]	1887	0.05934	0.05391	0.06036	0.05839	0.05421	0.06108	0.1026	0.05439	0.06119	0.006129	0.005014	0.006431
alpha	[+25, +35]	2187	0.05968	0.05564	0.06082	0.05793	0.05384	0.06056	0.1063	0.05551	0.06136	0.006745	0.004815	0.007053
alpha	[+35, +45]	2816	0.05372	0.05562	0.05521	0.05049	0.05604	0.05455	0.0964	0.05538	0.05574	0.00386	0.005175	0.003486
alpha	[+45, +50]	1762	0.05412	0.05282	0.05127	0.05082	0.05223	0.05057	0.08764	0.05505	0.05457	0.004696	0.004539	0.004363
alpha	[+50, +55]	2251	0.05054	0.05486	0.05068	0.04832	0.0532	0.05009	0.08405	0.05648	0.05163	0.003987	0.005376	0.004553
alpha	[+55, +60]	2783	0.04894	0.05475	0.04924	0.04503	0.05457	0.04751	0.09268	0.05557	0.04985	0.00362	0.005487	0.003614
alpha	[+60, +65]	3918	0.04703	0.04774	0.04659	0.04403	0.0477	0.04694	0.08495	0.05102	0.04843	0.005191	0.003901	0.003809
alpha	[+65, +70]	5564	0.04426	0.04175	0.04418	0.04132	0.04126	0.04387	0.08243	0.04375	0.04655	0.00469	0.005374	0.004181
alpha	[+70, +75]	8941	0.03991	0.04073	0.03923	0.03693	0.04083	0.03893	0.0771	0.04305	0.04152	0.004646	0.005596	0.005812
alpha	[+75, +80]	15855	0.03741	0.03764	0.03752	0.0343	0.03757	0.03745	0.07877	0.03937	0.03923	0.004742	0.004179	0.004693
alpha	[+80, +82]	3627	0.03645	0.03856	0.03821	0.03182	0.03843	0.03799	0.08463	0.04026	0.04007	0.002815	0.004434	0.003241
alpha	all	108222	0.04418	0.04407	0.04406	0.0405	0.04312	0.04296	0.08568	0.04677	0.0471	0.004568	0.004631	0.00468
beta	[-82, -80]	3635	0.1412	0.144	0.1386	0.1506	0.1458	0.1507	0.2001	0.1442	0.151	0.06001	0.02688	0.06059
beta	[-80, -75]	16269	0.1437	0.1401	0.1413	0.1431	0.1437	0.1441	0.1899	0.1441	0.1439	0.02032	0.02656	0.02122
beta	[-75, -70]	8905	0.1375	0.1355	0.1387	0.1356	0.1358	0.1394	0.2142	0.1382	0.1414	0.02119	0.01851	0.02228
beta	[-70, -65]	5574	0.1252	0.1359	0.1277	0.1203	0.1385	0.1302	0.1804	0.1377	0.1289	0.0141	0.01803	0.01373
beta	[-65, -60]	3715	0.137	0.1613	0.1363	0.1326	0.1605	0.1366	0.2431	0.1689	0.1419	0.01178	0.02969	0.01117

Dump closure per cotBeta band [cot]: q-width, core RMS, full RMS, median (2)

angle	beta_band_deg	N	dump_q_width	v5_q_width	v6_q_width	dump_core_rms	v5_core_rms	v6_core_rms	dump_full_rms	v5_full_rms	v6_full_rms	dump_median	v5_median	v6_median
beta	[-60, -55]	2763	0.1482	0.1842	0.1492	0.1308	0.1892	0.1478	0.1986	0.1947	0.1528	0.005346	0.04382	0.007284
beta	[-55, -50]	2220	0.1492	0.1837	0.147	0.1435	0.1857	0.1479	0.1986	0.1895	0.1573	0.04496	0.04208	0.04256
beta	[-50, -45]	1742	0.1554	0.1829	0.1558	0.1256	0.1918	0.1576	0.195	0.1952	0.162	0.01472	0.05294	0.027
beta	[-45, -35]	2819	0.2059	0.1821	0.2026	0.2118	0.1843	0.2149	0.2364	0.1908	0.2229	0.07023	0.04686	0.07494
beta	[-35, -25]	2191	0.1789	0.1807	0.178	0.157	0.1877	0.1799	0.2091	0.1891	0.1854	0.04288	0.04047	0.04081
beta	[-25, -15]	1871	0.2965	0.1775	0.2816	0.3073	0.1847	0.3045	0.2775	0.1848	0.3001	0.09075	0.05209	0.09758
beta	[-15, +0]	2489	0.2304	0.181	0.2234	0.1573	0.1799	0.2369	0.2382	0.1882	0.2406	0.08129	0.04102	0.08251
beta	[+0, +15]	2438	0.2145	0.1745	0.2095	0.1517	0.1777	0.2235	0.2285	0.1869	0.2312	-0.07279	-0.04749	-0.07669
beta	[+15, +25]	1887	0.3059	0.17	0.2961	0.334	0.1761	0.3101	0.2936	0.1802	0.3135	-0.08889	-0.04923	-0.08832
beta	[+25, +35]	2187	0.1924	0.1742	0.186	0.1601	0.1798	0.1922	0.2338	0.1837	0.1925	-0.03714	-0.05106	-0.04305
beta	[+35, +45]	2816	0.1893	0.175	0.1831	0.1833	0.1782	0.1855	0.2459	0.1855	0.2043	-0.05766	-0.04004	-0.05231
beta	[+45, +50]	1762	0.1475	0.1813	0.1473	0.1216	0.1893	0.1487	0.2365	0.1877	0.1585	-0.03356	-0.05117	-0.03413
beta	[+50, +55]	2251	0.1394	0.1725	0.1332	0.1368	0.1775	0.1439	0.2397	0.1832	0.1542	-0.05679	-0.05115	-0.06074
beta	[+55, +60]	2783	0.1408	0.1735	0.1366	0.1238	0.1785	0.1372	0.2443	0.1821	0.1445	-0.01857	-0.03931	-0.01344
beta	[+60, +65]	3918	0.1455	0.16	0.1469	0.1414	0.1627	0.1467	0.2503	0.1671	0.1528	-0.02472	-0.02844	-0.02337
beta	[+65, +70]	5564	0.1339	0.1401	0.1343	0.1275	0.1404	0.1353	0.3093	0.1428	0.1372	-0.008844	-0.01041	-0.006716
beta	[+70, +75]	8941	0.1388	0.1404	0.1391	0.1341	0.1401	0.1395	0.2804	0.1427	0.1411	-0.01262	-0.0126	-0.01308
beta	[+75, +80]	15855	0.1457	0.144	0.1455	0.1448	0.1486	0.1491	0.2721	0.1491	0.1493	-0.02731	-0.03215	-0.02811
beta	[+80, +82]	3627	0.1397	0.1467	0.1417	0.1518	0.1526	0.1566	0.2062	0.1513	0.1557	-0.06224	-0.03612	-0.06265
beta	all	108222	0.1548	0.1552	0.1543	0.1468	0.1537	0.1509	0.2379	0.1599	0.1649	-0.00123	-0.0004177	-0.0004386

Dump closure: pulls (dump residual - bias) / sigma (1)

angle	beta_band_deg	N	v5_pull_q_width	v6_pull_q_width	v5_pull_median	v6_pull_median
alpha	[-82, -80]	3635	0.9462	0.9933	0.04895	-2.188e-06
alpha	[-80, -75]	16269	0.9989	0.9956	-0.01089	3.806e-06
alpha	[-75, -70]	8905	0.9587	0.9902	-0.006294	3.802e-07
alpha	[-70, -65]	5574	1.044	0.997	0.01204	-3.213e-06
alpha	[-65, -60]	3715	0.9941	0.9926	0.001292	-1.187e-07
alpha	[-60, -55]	2763	0.8991	1.009	-0.02274	-2.496e-06
alpha	[-55, -50]	2220	0.952	0.9904	-0.07581	4.539e-06
alpha	[-50, -45]	1742	0.9633	0.9988	0.0333	4.931e-06
alpha	[-45, -35]	2819	1.003	1.009	0.01598	-4.487e-06
alpha	[-35, -25]	2191	1.035	1.007	0.03977	8.995e-07
alpha	[-25, -15]	1871	1.058	0.998	-0.01312	6.578e-07
alpha	[-15, +0]	2489	1.162	1.005	0.02073	-3.621e-06
alpha	[+0, +15]	2438	1.174	1.003	0.06488	-2.766e-07
alpha	[+15, +25]	1887	1.107	0.9788	0.02652	-1.209e-06
alpha	[+25, +35]	2187	1.09	0.9917	0.01191	6.816e-06
alpha	[+35, +45]	2816	0.9731	0.9874	-0.03475	4.488e-06
alpha	[+45, +50]	1762	0.9767	0.9979	-0.02042	7.614e-07
alpha	[+50, +55]	2251	0.9123	1.002	-0.01197	-5.006e-06
alpha	[+55, +60]	2783	0.8905	0.9979	-0.03086	5.497e-06
alpha	[+60, +65]	3918	0.9621	0.99	0.006879	-1.707e-06
alpha	[+65, +70]	5564	1.066	0.994	-0.003806	6.963e-07
alpha	[+70, +75]	8941	0.9546	1	0.0027	-5.089e-06
alpha	[+75, +80]	15855	0.9923	0.9927	0.01149	3.581e-07
alpha	[+80, +82]	3627	0.9708	0.9867	-0.04021	7.985e-08
alpha	all	108222	0.9934	0.9963	6.138e-07	-5.824e-08
beta	[-82, -80]	3635	0.9849	1.007	0.2286	9.988e-07
beta	[-80, -75]	16269	1.012	1.003	-0.03862	2.45e-07
beta	[-75, -70]	8905	1.02	1.001	0.02568	-8.59e-07
beta	[-70, -65]	5574	0.9426	1.008	-0.02157	2.381e-07
beta	[-65, -60]	3715	0.8683	1.008	-0.1145	5.242e-07

Dump closure: pulls (dump residual - bias) / sigma (2)

angle	beta_band_deg	N	v5_pull_q_width	v6_pull_q_width	v5_pull_median	v6_pull_median
beta	[-60, -55]	2763	0.8175	1.015	-0.1946	-5.937e-08
beta	[-55, -50]	2220	0.8046	1.002	0.01322	-7.367e-07
beta	[-50, -45]	1742	0.8576	1.016	-0.1397	-7.02e-07
beta	[-45, -35]	2819	1.12	0.9999	0.1576	-1.288e-06
beta	[-35, -25]	2191	0.9627	1.003	0.004858	6.539e-07
beta	[-25, -15]	1871	1.62	1.002	0.2544	-1.64e-07
beta	[-15, +0]	2489	1.246	1.011	0.2103	9.436e-08
beta	[+0, +15]	2438	1.231	0.9989	-0.1464	6.003e-07
beta	[+15, +25]	1887	1.772	1.005	-0.2052	6.603e-07
beta	[+25, +35]	2187	1.088	1	0.05321	-7.199e-07
beta	[+35, +45]	2816	1.061	0.9987	-0.06393	1.2e-06
beta	[+45, +50]	1762	0.8187	1.019	0.07907	-4.132e-07
beta	[+50, +55]	2251	0.7797	1.002	-0.06343	-1.119e-06
beta	[+55, +60]	2783	0.8103	0.9945	0.1519	2.467e-06
beta	[+60, +65]	3918	0.8969	0.9992	0.04335	-2.427e-06
beta	[+65, +70]	5564	0.9595	1.006	0.02286	6.594e-07
beta	[+70, +75]	8941	0.9947	1.001	-0.004517	1.4e-07
beta	[+75, +80]	15855	1.014	1.001	0.03244	-7.089e-07
beta	[+80, +82]	3627	0.9621	1.004	-0.1982	-1.661e-07
beta	all	108222	1	0.9877	-7.699e-07	4.261e-08

Refit v5 vs v6: residual q68 half-widths and medians vs the TP; chi2; selected-hit purity (median column)

run	quantity	tp_pt	n	q68_halfwidth	median	q90	q99	frac_gt_100
v5	d0 [um]	[0, inf)	16303	60.22	0.008241	nan	nan	nan
v5	d0 [um]	[2, 5)	13309	56.94	0.03327	nan	nan	nan
v5	d0 [um]	[5, inf)	1666	25.25	0.3903	nan	nan	nan
v5	z0 [um]	[0, inf)	16303	157.8	0.2301	nan	nan	nan
v5	z0 [um]	[2, 5)	13309	142.5	0.2718	nan	nan	nan
v5	z0 [um]	[5, inf)	1666	57.95	0.2813	nan	nan	nan
v5	pT (reco - true) / true	[0, inf)	16303	0.01959	0.001407	nan	nan	nan
v5	pT (reco - true) / true	[2, 5)	13309	0.01829	0.0005426	nan	nan	nan
v5	pT (reco - true) / true	[5, inf)	1666	0.01858	0.001878	nan	nan	nan
v5	chi2 IT total	all	16303	nan	32.15	830.3	1.648e+04	0.2325
v5	chi2 alpha	all	16303	nan	2.99	14.95	1017	0.03349
v5	chi2 beta	all	16303	nan	3.668	152.4	1300	0.1251
v5	selected-hit purity (own TP)	all	46518	nan	0.8548	nan	nan	nan
v6	d0 [um]	[0, inf)	16303	60.22	-0.04534	nan	nan	nan
v6	d0 [um]	[2, 5)	13309	56.98	-0.01293	nan	nan	nan
v6	d0 [um]	[5, inf)	1666	25.29	0.4287	nan	nan	nan
v6	z0 [um]	[0, inf)	16303	158.1	0.1431	nan	nan	nan
v6	z0 [um]	[2, 5)	13309	142.5	0.1621	nan	nan	nan
v6	z0 [um]	[5, inf)	1666	57.6	0.2563	nan	nan	nan
v6	pT (reco - true) / true	[0, inf)	16303	0.0196	0.001389	nan	nan	nan
v6	pT (reco - true) / true	[2, 5)	13309	0.01829	0.0005467	nan	nan	nan
v6	pT (reco - true) / true	[5, inf)	1666	0.01854	0.001892	nan	nan	nan
v6	chi2 IT total	all	16303	nan	32.98	828.7	1.655e+04	0.2318
v6	chi2 alpha	all	16303	nan	3.051	15.09	964.9	0.03226
v6	chi2 beta	all	16303	nan	3.713	146.3	1457	0.123
v6	selected-hit purity (own TP)	all	46520	nan	0.8548	nan	nan	nan

Nano checks

check	value
branches compared	862
identical	696
differing SmartPixels	166
differing non-SmartPixels	0
non-finite values in v6 SmartPixels float branches	0
file bytes v5 / v6	139024 / 343160